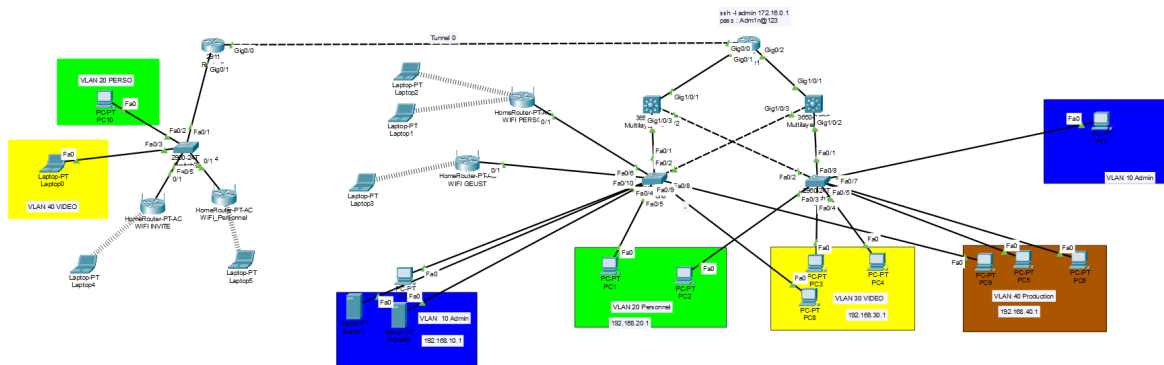


DEMO — SAE 3.3

Infrastructure réseau multi-sites (Siège / Succursale)



1. Vérification des VLAN et des trunks (Siège)

Sur **CORE-SW1** :

```
show vlan brief
show interfaces trunk
```

- VLAN présents : 10 (ADMIN), 20 (PERSONNEL), 30 (VIDEO), 40 (PRODUCTION), 50 (GUEST)
- Trunks actifs entre CORE-SW1, CORE-SW2 et switchs d'accès

```
CORE-SW1#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Gig1/0/4, Gig1/0/5, Gig1/0/6, Gig1/0/7 Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11 Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15 Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19 Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23 Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
10 ADMIN	active	
20 PERSONNEL	active	
30 VIDEO	active	
40 PRODUCTION	active	
50 VLAN0050	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
CORE-SW1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig1/0/2	on	802.1q	trunking	1
Gig1/0/3	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig1/0/2	10,20,30,40,50
Gig1/0/3	10,20,30,40,50

Port	Vlans allowed and active in management domain
Gig1/0/2	10,20,30,40,50
Gig1/0/3	10,20,30,40,50

Port	Vlans in spanning tree forwarding state and not pruned
Gig1/0/2	10,20,30,40,50
Gig1/0/3	20,50

2. Haute disponibilité — HSRP

Sur **CORE-SW1** et **CORE-SW2** :

```
show standby brief
```

- CORE-SW1 = Active
- CORE-SW2 = Standby
- Passerelle virtuelle en **.1** pour chaque VLAN

```
CORE-SW2#show standby brief
```

```

      P indicates configured to preempt.
      |
Interface    Grp  Pri P State    Active        Standby        Virtual IP
Vl10         10   100 P Standby   192.168.10.2   local          192.168.10.1
Vl20         20   100 P Standby   192.168.20.2   local          192.168.20.1
Vl30         30   100 P Standby   192.168.30.2   local          192.168.30.1
Vl40         40   100 P Standby   192.168.40.2   local          192.168.40.1
Vl50         50   100 P Standby   192.168.50.2   local          192.168.50.1

```

```

CORE-SW1#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri P State      Active      Standby      Virtual IP
Vl10           10   110 P Active     local       192.168.10.3  192.168.10.1
Vl20           20   110 P Active     local       192.168.20.2  192.168.20.1
Vl30           30   110 P Active     local       192.168.30.2  192.168.30.1
Vl40           40   110 P Active     local       192.168.40.2  192.168.40.1
Vl50           50   110 P Active     local       192.168.50.2  192.168.50.1
CORE-SW1#

```

3. Routage dynamique — OSPF

Siège

Sur **CORE-SW1** :

```
show ip ospf neighbor
```

- Voisins OSPF établis avec R-HQ et CORE-SW2

```
CORE-SW1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:31	172.16.0.1	GigabitEthernet1/0/1

```
CORE-SW1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:37	172.16.0.1	GigabitEthernet1/0/1

```

CORE-SW1#show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 6
    Area has no authentication
    SPF algorithm executed 5 times
    Area ranges are
    Number of LSA 6. Checksum Sum 0x038035
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

Succursale

Sur **R-BRANCH** :

```
show ip route
```

- Routes vers les réseaux du siège apprises via OSPF
- Route par défaut vers le tunnel GRE

```
R-BRANCH# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Tunnel0
L       10.0.0.1/32 is directly connected, Tunnel0
C       10.255.255.1/32 is directly connected, Loopback10
    172.16.0.0/30 is subnetted, 2 subnets
O       172.16.0.0/30 [110/1001] via 10.0.0.2, 00:32:02, Tunnel0
O       172.16.0.4/30 [110/1001] via 10.0.0.2, 01:24:52, Tunnel0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/30 is directly connected, GigabitEthernet0/0
L       192.168.3.2/32 is directly connected, GigabitEthernet0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.10.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
C       192.168.10.253/32 is directly connected, Loopback0
O       192.168.20.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.30.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.40.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.50.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
    192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/1.20
L       192.168.120.254/32 is directly connected, GigabitEthernet0/1.20
    192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.140.0/24 is directly connected, GigabitEthernet0/1.40
L       192.168.140.254/32 is directly connected, GigabitEthernet0/1.40
    192.168.150.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.150.0/24 is directly connected, GigabitEthernet0/1.50
L       192.168.150.254/32 is directly connected, GigabitEthernet0/1.50
S*    0.0.0.0/0 [1/0] via 10.0.0.2
```

4. Interconnexion sécurisée — Tunnel GRE

Sur R-HQ :

```
show interface tunnel0
```

```

R-HQ#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.0.0.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.168.3.1 (GigabitEthernet0/0), destination 192.168.3.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 56 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    726 packets input, 53840 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

Sur **R-BRANCH** :

```
show interface tunnel0
```

```

R-BRANCH#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.0.0.1/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.168.3.2 (GigabitEthernet0/0), destination 192.168.3.1
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 57 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    689 packets input, 55530 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

- Tunnel GRE **UP / UP**
- Trafic inter-sites fonctionnel

5. DHCP centralisé (Siège)

Sur un **PC du VLAN 20 (PERSONNEL – Siège)** :

- Desktop → IP Configuration → **DHCP**
- IP reçue dynamiquement
- DNS = **192.168.10.10**
- Passerelle = **192.168.20.1**

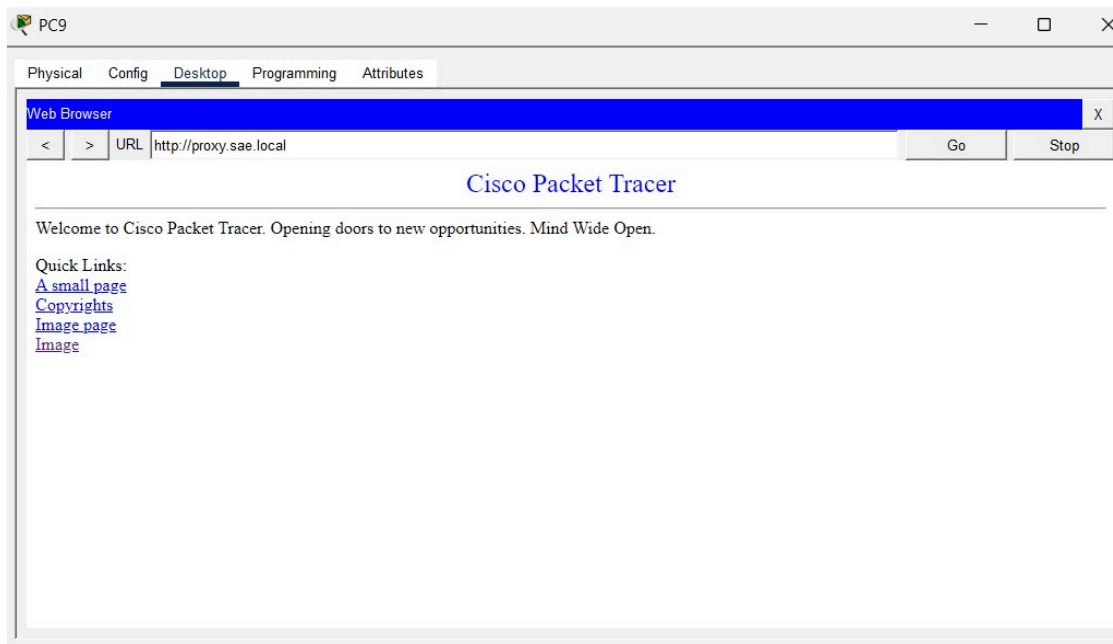
(Le DHCP est centralisé sur le serveur du siège)

IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IPv4 Address	192.168.20.135
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	192.168.10.10

6. Proxy Web (Siège)

Depuis un PC interne :

- Navigateur → `http://proxy.sae.local`



- ou `http://192.168.10.12`

Accès Web via le proxy HTTP centralisé

7. Sécurité inter-VLAN (ACL)

Exemple :

- VLAN 20 (PERSONNEL) vers VLAN 40 (PRODUCTION)

Depuis un PC VLAN 20 :

```
ping 192.168.40.1
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping bloqué : ACL fonctionnelle

8. Wi-Fi invité (VLAN 50) — Isolation réseau

Siège

Sur **CORE-SW1** :

```
show ip interface vlan 50
```

- ACL **GUEST-WEB** appliquée
- DHCP autorisé
- Accès Web uniquement via proxy

Depuis un **client WiFi invité** :

- Accès à **http://192.168.10.12**



- Non-Accès aux VLAN internes (ADMIN / PERSONNEL / PROD)

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.50.2: Destination host unreachable.
Reply from 192.168.50.2: Destination host unreachable.
Reply from 192.168.50.2: Destination host unreachable.
Reply from 192.168.50.2: Destination host unreachable.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

9. Wi-Fi invité — Succursale

Sur **R-BRANCH** :

```
show ip interface g0/1.50
```

```
R-BRANCH#show ip interface g0/1.50
GigabitEthernet0/1.50 is up, line protocol is up (connected)
  Internet address is 192.168.150.254/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 192.168.10.10
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is GUEST-WEB
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
```

- VLAN invité isolé

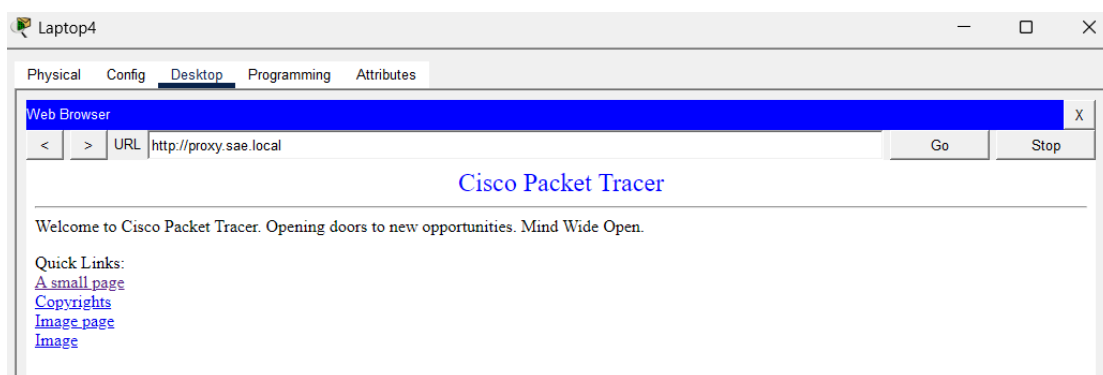
```
C:\>ping 192.168.120.1

Pinging 192.168.120.1 with 32 bytes of data:

Reply from 192.168.150.254: Destination host unreachable.
Reply from 192.168.150.254: Destination host unreachable.
Reply from 192.168.150.254: Destination host unreachable.
Reply from 192.168.150.254: Destination host unreachable.

Ping statistics for 192.168.120.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Accès Web via proxy du siège



- Accès réseaux internes bloqué

(En raison d'une limitation connue de Packet Tracer avec DHCP relay via tunnel GRE, la démonstration peut être faite en IP statique côté succursale.)

10. Accès distant sécurisé — SSH

Depuis un PC **VLAN 10 (ADMIN)** :

```
ssh -l admin core-sw1.sae.local
```

```
C:\>ssh -l admin core-sw1.sae.local

Trying 192.168.10.2 ...
Password:

CORE-SW1#sh
CORE-SW1#show vl
CORE-SW1#show vlan br
```

VLAN Name	Status	Ports
1 default	active	Gig1/0/4, Gig1/0/5, Gig1/0/6, Gig1/0/7 Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11 Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15 Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19 Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23 Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
10 ADMIN	active	
20 PERSONNEL	active	
30 VIDEO	active	
40 PRODUCTION	active	
50 VLAN0050	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
CORE-SW1#
```

Accès SSH autorisé

Depuis un autre VLAN :

```
C:\>ssh -l admin core-sw1.sae.local

Trying 192.168.10.2 ...% Connection refused by remote host
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:3EFF:FE62:B684
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.20.56
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.20.1
```

Accès refusé (ACL de management)

11. Centralisation des logs (Syslog)

Sur **CORE-SW1** :

```
show logging
```

```

CORE-SW1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                    0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 34 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 34 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 34 message lines logged
  Logging to 192.168.10.11 (udp port 514, audit disabled,
    authentication disabled, encryption disabled, link up),
    34 message lines logged,
    0 message lines rate-limited,
    0 message lines dropped-by-MD,
    xml disabled, sequence number disabled
    filtering disabled

Log Buffer (4096 bytes):

%LINK-3-UPDOWN: Interface Vlan30, changed state to down

```

Logs envoyés vers 192.168.10.11

Supervision centralisée

Conclusion

- VLAN segmentés
- Haute disponibilité (HSRP)
- Routage dynamique (OSPF)
- Interconnexion sécurisée (GRE)
- DHCP centralisé
- Proxy Web
- Wi-Fi invité isolé
- Sécurité inter-VLAN
- Administration sécurisée (SSH)

Le cahier des charges est respecté.